Float and Sink: One Leaf, Both Processes

Lab · Canvas Original #41 · Module 3, Lesson 3 · LT 3.2 (rate of photosynthesis) and LT 3.4 (how photosynthesis and respiration connect)

Name _____ Period _____ Date _____ Role: timer · counter · recorder · materials (circle one)

Question. A leaf in the light gives off oxygen. Does the same leaf also use oxygen, and how would you see it?
A spinach leaf is full of air spaces. Pull the air out with a syringe and fill the spaces with a little baking-soda water (it supplies carbon dioxide), and a small disk of leaf sinks. In light, its chloroplasts make sugar and release oxygen; the oxygen fills the spaces again and the disk rises. Every living leaf cell also carries out cellular respiration, day and night, using oxygen.

Safety and materials

Goggles on. The syringe has no needle; keep a finger over the tip while you pull. The solution is baking soda, water and one drop of dish soap, and it goes down the sink when you finish.

Per group: fresh spinach leaves, a straw or one-hole punch, a 10–60 mL plastic syringe, a cup of baking-soda solution (about 1/8 teaspoon in 300 mL water plus 1 drop of soap), three clear cups labeled A, B and C, a lamp, a ruler, the class dark box, a timer.

Part 1 · Sink the disks (10 minutes)

1. Punch 30 disks from the leaf, avoiding the big veins.
2. Pull the plunger out, drop the disks into the syringe barrel, push the plunger back in to the 5 mL mark, and draw up about 10 mL of solution.
3. Point the tip up, push out the air, then seal the tip with a finger and pull the plunger back hard for 10 seconds. Bubbles leave the disks.
4. Release, tap, and repeat until most disks sink. Pour them out and split them: 10 disks in each cup, covered by about 3 cm of solution.

Part 2 · Light near, light far, no light (12 minutes)

Cup A sits 10 cm from the lamp. Cup B sits 30 cm from the lamp. Cup C goes in the dark box. Start the timer when the lamp goes on, and count the floating disks in A and B every minute; peek into C only at 6 and 12 minutes.

Predict before you start: which cup will have half its disks floating first, and which may never get there?

_____

Minute	0	1	2	3	4	5	6	7	8	9	10	11	12
A · 10 cm													
B · 30 cm													
C · dark													

Number of disks floating (out of 10). Grey boxes: leave cup C closed.

Part 3 · Lights out for cup A (rest of class, then next class)

When at least 8 disks in cup A float, move cup A into the dark box. Count its floating disks every 2 minutes for 10 minutes, then once more at the end of class.

Predict: what will the floating disks in cup A do in the dark, and why? _____

Minutes in the dark	0	2	4	6	8	10	End of class
Cup A disks floating							

Part 4 · Pool the class data

For each cup, record the minute when 5 of the 10 disks were floating (the half-float time). Write “never” if it did not happen in 12 minutes. Copy your numbers to the board, then copy the class average.

	Our half-float time (min)	Class average (min)	Groups that never reached half
A · 10 cm			
B · 30 cm			
C · dark			

Graph the floating disks against time for cups A and B on the grid your teacher provides, or on GoodNotes: time on the x-axis, disks floating on the y-axis, a key for each line.

Part 5 · Make sense of it

1. What made the disks in cup A rise? Name the gas, the process and the organelle.

_____

_____

2. Compare cups A and B. What does the difference tell you about light and the rate of photosynthesis? Use your half-float times.

_____

_____

3. Cup C sat in the same solution with the same carbon dioxide. Why did its disks stay down?

_____

_____

4. In Part 3, what happened to cup A's disks in the dark? Name the process that used the oxygen, and the organelle where it happens.

_____

_____

5. Complete both lines. In the light, the leaf disk makes oxygen (faster / slower) than it uses it. In the dark, the disk (makes / uses) oxygen only.

6. Matter cycles and energy flows. Name one atom that cycled through this lab, and trace where the energy went from the lamp onward.

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_____

Advanced and P3: design your own run. Change one variable (distance, baking soda amount, a green or red filter over the lamp, or a different leaf), keep everything else the same, and predict before you test. Variable changed: _____ Controlled: _____

Take it further: read **Dawn in Cobbs Creek** (Biology #20, scienceonepagers.org), where the same day-and-night pattern plays out in a whole creek. Absent on lab day? Run the leaf-disk simulation in Biochemistrypedia, “Capturing Sunlight” (Bio Tool #30).

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